

Table 3: Spectral sparsity — optimizer × objective (FP32)

	GRPO (RL)	SFT
AdamW	rank@90% = 190 stable rank = 5.4	rank@90% = 450 stable rank = 10.6
Muon	rank@90% = 752 stable rank = 330.1	rank@90% = 794 stable rank = 302.9

AdamW+GRPO: 4× more concentrated than AdamW+SFT, 61× more than Muon+GRPO

Muon is indifferent to objective (stable rank $\approx$ 300 in both RL and SFT)

Max possible rank = 1536 • FP32 training • ~1.1M tokens • Qwen2-1.5B